Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/06/2010
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using a Bed Rest Analog-NNX08BA58G		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2008	**End Date:**	02/29/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date will be 2/29/2012 per C. Guidry/JSC (Ed., 8/9/2011) NOTE: End date is now 1/15/2013 per JSC HRP Master Task List 5/20/2011 (Ed., 7/21/2011) NOTE: Received funding augmentation/new end date to 8/31/2011 (per NSSC info 2/2010) NOTE: Received NCE to 12/31/2010 per JSC (8/09) NOTE: Received NCE to 7/31/2009 (per NSSC info; added 8/09) NOTE: Received NCE to 4/30/2009; original end date was 12/31/2008 (per NSSC info; added 8/09)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Licata, Angelo (Cleveland Clinic) Ilaslan, Hakan (Cleveland Clinic)		
Grant/Contract No.:	NNX08BA58G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involves quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on DLS theory. The specific aim of this project is to determine if individualized mechanical load replacement and supplementation during bedrest can serve as a countermeasure to losses in BMD, bone quality, and muscle atrophy during 12 weeks of bedrest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	The total number of subjects who have completed the study at the Cleveland Clinic is now 11 (6 controls and 5 exercisers). During the last year, the administrative and regulatory procedures needed to transfer this study to the NASA Flight Analogs Research Unit (FARU) have been completed. The study has been successfully relocated to the NASA Flight Analogs Research Unit at the University of Texas Medical Branch in Galveston, TX for completion of the remaining subjects. Four subjects began screening activities for the Texas arm of the study. Three of these subjects were withdrawn or disqualified from screening as they did not meet the inclusion requirements. One subject was enrolled in the full study. This subject completed the pre-bed rest phase and five days of bed rest. On the fifth day of bed rest, the subject was withdrawn from the study at the request of the NASA Committee for the Protection of Human Subjects (CPHS) pending the review of ocular test results from other head-down bed rest studies. The project team has modified existing mathematical theories on daily load stimulus (DLS) incorporating new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the effects of standing. This revised algorithm has been dubbed the enhanced DLS (EDLS). Extensive MATLAB code has been written to allow users to analyze complete sets of in-shoe force data and calculate EDLS. The EDLS is being used to formulate the individualized exercise prescriptions used by the exercising subjects. The EDLS algorithm has been documented and published in the journal "Aviation, Space, and Environmental Medicine."
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Genc KO, Cavanagh PR. "Examining the Changes in Femoral Strength due to Long Duration Space Flight: A Preliminary Modeling Framework using a Bedrest Analog." Presented at the Northwest Biomechanics Symposium, Seattle, WA, May 21-22, 2010. Proceedings from the Northwest Biomechanics Symposium, Seattle, WA, May 21-22, 2010. , May-2010
Abstracts for Journals and Proceedings	Genc KO, Gopalakrishnan R, Kuklis MM, Maender CC, Rice AJ, Cavanagh, PR. "Using the enhanced daily load stimulus model to quantify the mechanical load and bone mineral density changes experienced by crew members on the International Space Station." Presented at the 56th Annual Meeting of the Orthopaedic Research Society, New Orleans, LA, March 6-9, 2010. Proceedings from the 56th Annual Meeting of the Orthopaedic Research Society, New Orleans, LA, March 6-9, 2010. , Mar-2010